# Experimental Design

**Research Objective:**

This study employs an LLM agent-based simulation to examine two interconnected questions: first, whether confirmation bias shapes the different adoption of politically slanted misinformation among ideologically distinct agent groups; and second, how the three-way interaction among rumour slant, agent political identity, and refutation source type jointly determines both individual belief change and population-level polarization. Of particular interest is whether AI-generated refutation can bypass the partisan attribution mechanisms that constrain the effectiveness of refutations with partisan labels.

## 1. Model Components

**Agents:** 30 LLM-powered agents with heterogeneous personas, each agent is equipped with a **dual-memory system** (comprising short-term and long-term memory) and a **reflective reasoning mechanism**, enabling them to integrate interaction information with their unique persona-driven traits to dynamically update their beliefs about misinformation in a human-like cognitive manner.

**Environment:** A dynamic interaction environment based on a **random mixing pool** of LLM-powered agents, where agents can randomly interact with each other with a daily frequency.

**Interaction:** During each simulated day (time step), every agent randomly selects **c** peers from the total agent pool to engage in textual opinion exchanges. While the frequency of interactions each pair of agents is fixed at **d** for each step to ensure computational consistency, the specific interaction partners are dynamically and randomly reassigned each day to replicate the fluid and stochastic nature of information flow in realistic social media.

## 2. Agent Architecture (Micro-level)

To make agents act as realistic as humans, ensuring "Human Alignment" and "Believability", each agent includes:

| Component | Implementation | Source/Theory |
|---|---|---|
| **Persona (Heterogeneity)** | Prompt-engineered profiles dictating age, education, name and different political ideology level (a 7 likert level between very liberal to very conservative). | (Nyhan & Reifler, 2010) |
| **Memory System** | The system is built on the premise that an individual's opinion is influenced by social interactions in a way that is gradual and cumulative rather than immediate. It utilizes two distinct memory types to mimic human-like thinking and reflection processes;<br><br>**Short-term memory:**<br><br>● Function: It reflects and summarizes interactions from the current day.<br>● Management: To manage the potentially vast volume of daily data, this memory is cleared at the end of each day after being processed.<br>● Prompting: Agents are prompted to "Summarize the opinions you have heard in a few sentences, including whether or not they believe in the news."<br><br>**Long-term memory:**<br><br>● Function: It compresses and stores a summarized history of all past interactions to provide broader context for the agent.<br>● Integration: It is updated daily by integrating the latest short-term | Liu et al. (2024) |

| | | |
|---|---|---|
| | summary with the existing long-term record.<br>● Prompting: The update process involves a prompt to "Recap previous long-term memory [and] today's short-term summary... ensuring to maintain continuity and add any new insights." | |
| **Cognitive Biases** | The system prompts agents to exhibit **Confirmation Bias** when evaluating news. | Shu et al. (2017) |
| **Reflection Loop** | **Decision-Making**: At the end of each day, the agent reflects on the interactions stored in these memories.<br>**Belief Evolution**: This reflection process allows the agent's opinion to evolve dynamically over time, striking a balance between retaining historical context and processing new social information. | Liu et al. (2024) |

This agent architecture equips each agent with persona $p_i$ including their name, age, education level and political ideology level, since these are variables that might influence their attitude toward fake news and different refutations. The primary objective of this study is to investigate how ideological congruence moderates the effectiveness of misinformation refutation. So agents are equally distributed across the 7 ideology levels, other demographic variables are randomly assigned across these groups to mitigate confounding effects on the dependent variable.

The dual-memory system comprises a **short-term memory** for summarizing daily interactions and a **long-term memory** for storing a compressed history of past exchanges.

This architecture enables agents to undergo a **reflective reasoning process**, where daily summaries are integrated into a continuous long-term context to drive human-like opinion dynamics.

This study construct a simulation with a pool of **35** LLM agents $A = (a_{1,}, \ldots, a_N)$ and a fake new topic F with a specific political slant. Just like the study of liu et al (2024), for initialization, each agent has a unique persona, including their initial attitude towards the fale news. On t-th day, each agent $a_i$ will randomly interact with $c$ other agents from the pool $A$. At the end of the day, every agent reflects on the exchanged information and decides whether to believe in the fake news. This process is iterated over $T$ days. Through these daily iterations, agent opinions are accumulated to plot the trajectory of different populations, resulting in a curve that illustrates the dynamics of the fake news within the network. Additionally, the evolution of the agents' beliefs would also be scrutinized to examine how individual and collective opinions shift over time (Liu et al, 2024).

## 3. Hypotheses to Test (The "What-If" Scenarios)

The following hypotheses are formulated within an exploratory direction. Due to the limited agent number in this study (30), and the novel application of LLM-based methods to partisan congruence research, these hypothesis are formed as directional expectations rather than confirmatory predictions. The primary aim of this study is to identify consistent patterns from repeated simulation runs and then can contribute to the design of larger-scale empirical studies for this field in the long run.

In this study, there are hypotheses in two stages:

### 3.1 Rumour Propagation Hypotheses

#### H1— Confirmation Bias in Rumour Adoption

Agents whose political identity is congruent with the ideological slant of the rumor are expected to adopt the rumor at a faster rate than ideologically incongruent agents, exhibiting comparatively higher initial belief intensity scores and shorter time-to-infection on average across simulation runs during the peer-to-peer propagation phase. This directional pattern is treated as the evidence of confirmation bias operated at the agent level.

## 3.2 Intervention Effectiveness Hypotheses

### H2 — Three-Way Political Identity Congruence Effect

The effectiveness of refutation interventions is jointly determined by the interaction among the rumor's ideological slant, the agent's political identity, and the refutation source's partisan affiliation.

- **H2a — Congruent Partisan Advantage**

When a partisan refutation source is ideologically aligned with the agent's political identity and opposed to the rumor's slant, it is expected to produce comparatively greater belief change (AAS) and higher RSR than the incongruent partisan condition, representing the extra refutation effect coming from identity proximity.

- **H2b — Conditional Partisan Backfire Effect**

The impact of ideologically incongruent partisan refutations is moderated by the alignment between the agent's identity and the rumor's slant. Specifically, a backfire effect—characterized by an increase or persistence in belief intensity—will occur only when the agent's political identity is congruent with the rumor's ideological slant. In contrast, when the agent's identity is incongruent with the rumor's slant, even refutations from an opposed partisan source will reinforce disbelief, leading to a decrease in belief intensity.

The empirical studies about backfire effect have got mixed results so far. Nyhan and Reifler's study found the empirical evidence of backfire effect on subjects with strongly conservative political stance when tring to correct their belief of the misinformation which is congruent with their predisposition (Nyhan & Reifler, 2010). While Wood and Porter found that evidence of factual backfire is far more tenuous than prior research suggests. They enrolled more than 10,100 subjects in their experiments and tested 52 issues of potential backfire, but found no supportive evidence of it despite testing precisely the kinds of polarized issues where backfire should be expected. Based on the vague results from previous empirical studies, H2b is treated as an exploratory observation rather than a confirmative directional prediction.

### H3 — Neutral Institutional Baseline Consistency

Neutral institutional refutations are expected to yield moderate but consistent belief change (AAS) across both Liberal and Conservative agent clusters compared to partisan conditions, as the absence of a political label reduces the influnce of identity-congruence level. Consequently, this condition is expected to produce a comparatively lower **Inter-group Belief Gap (IBG)** and **Polarization Index (PI)** relative to partisan source conditions.

**H4 — AI-Driven Cognitive Alignment and Cross-Partisan Penetration**

AI-generated refutations are expected to exhibit lower variance in effectiveness across Liberal and Conservative agents compared to partisan conditions, resulting from the absence of partisan attribution cues. Jia et al indicated that **conservatives tended to view algorithmic labels as more objective and politically unbiased than third-party neutral labels,** suggesting that algorithmic labelling may serve as a viable alternative or complement to traditional third-party fact-checking, particularly for politically congruent misinformation targeting conservative audiences (Jia et al., 2022). Whether AI refutation outperforms neutral institutional refutation in overall RSR and AAS especially on conservative groups is treated as an open question to be explored in this study.

# 4. Experimental Variables

This study will test the following intervention strategies against a **Control Group** (no interventions, pure organic spread).

The interventions this study aims to explore are as below:

- **Authoritative media's rumour refutation notice and explanation with different political stance**
- **Authoritative third-party fack checker's rumour refutation notice and explanation with neutral political stance.**
- **AI generated rumour refutation notice and explanation.**

## 4.1 Independent Variable:

The independent variables represent the factors manipulated to observe their impact on the agents' cognitive shifts.

- **Source Type:**

This categorical variable defines the nature of the corrective information provided to the agents. It consists of four levels:

- ➢ Control: No intervention is provided.
- ➢ Partisan Official Refutation: Fact-checks issued by politically labeled authoritative media (Liberal or Conservative).
- ➢ Neutral Institutional Refutation: Fact-checks from non-partisan, high-authority entities (e.g., a global fact-check union).
- ➢ AI-Generated Explanation: Fact-checks from an advanced LLM without an explicit political persona.

- **Source-Stance Congruence:**

Source-Stance Congruence: This variable measures the ideological alignment between the agent's persona and the intervention source. It is operationalized into three conditions:

- ➢ Congruent: The source entity's ideology matches the agent's (e.g., Conservative agent receiving a Conservative media refutation).
- ➢ Incongruent: The source entity's ideology opposes the agent's (e.g., Conservative agent receiving a Liberal media refutation).
- ➢ Neutral: Applied to both the Neutral Institutional and AI-Generated conditions, where the refutation carries no partisan identifier.

## 4.2 Dependent Variables：Evaluation Metrics

From the micro level, liu et al adopted a modified SIR (Susceptible-Infectious-Recovered) model, where agents can transition between being susceptible to fake news, becoming infected by interacting with other infected agents, and then being considered recovered after the misinformation is corrected. And recovered agents can become infected again due to the dynamic nature of people's opinions (Liu et al, 2024). The micro state of agents transition like this direction: $S \rightarrow I \rightleftharpoons R$. And unlike previous studies, the state transition of agents doesn't change by mere numerical calculation, but by using text descriptions to simulate each individual's perspective on fake news after reasoning process, providing a richer and more nuanced explanation.

So the dependent variables track both the agents' state transitions by their responses and the overall population dynamics resulting from the interventions.

**Micro-level:**

- ➢ **Belief Intensity (Credibility Score) — Original observed value (Baseline Indicator):** A continuous numerical metric (e.g., scaled from 1 to 7) representing the agent's subjective level of certainty regarding the rumor's veracity. This allows for a more granular statistical analysis of "marginal" attitude shifts.
- ➢ **Attitude Shift Magnitude — Dynamic change value (Process Indicator):** The calculated difference between an agent's pre-intervention belief score and their post-intervention score, quantifying the persuasive efficacy of the debunking message.
- ➢ **Final Belief State — Threshold classification value (qualitative indicator):** A categorical outcome for each agent at the end of the simulation, following the SIR (Susceptible-Infected-Recovered) model logic. Agents are classified as either **Infected** (still believing the rumor) or **Recovered** (successfully refuted). A threshold is typically established (e.g., Score >= 5) to classify an agent as **Infected**, whereas those below it are categorized as **Recovered**. This process effectively transforms continuous numerical data into discrete **identity labels**.

**Macro–level Indicators (Population & Cluster Analysis)**:

- **Refutation Success Rate (RSR):** The aggregate percentage of the total agent population that successfully transitioned from an "Infected" state to a "Recovered" state across different experimental groups.
- **Average Attitude Shift (AAS)**: The mean magnitude of change in belief intensity across the population, reflecting the collective persuasive impact of the intervention.
- **Polarization Index (PI)**: The standard deviation of belief scores among all agents, used to quantify the degree of ideological divergence or consensus-building following the intervention.

- **Inter-group Belief Gap (IBG)**: The absolute difference in mean belief scores between Liberal and Conservative agent clusters, measuring the intervention's efficacy in bridging partisan divides.

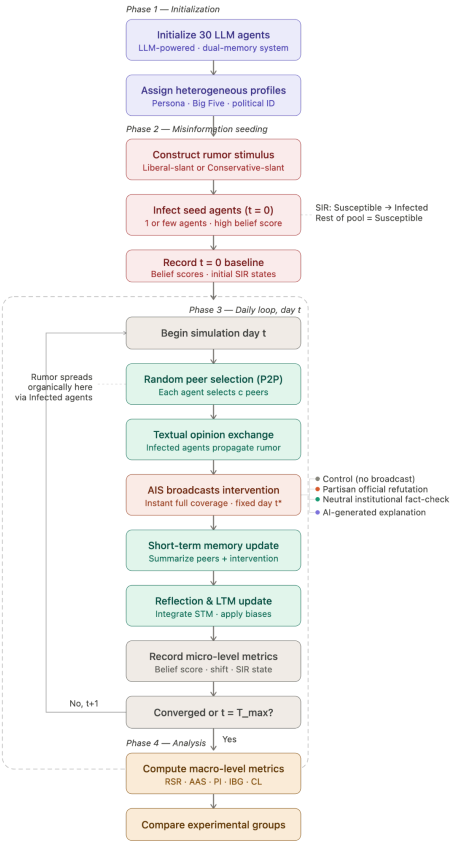
### 4.3 Moderating Variables (Control Variables)

These are intrinsic traits that remain constant during the simulation but may influence the relationship between the IVs and DVs.

- **Political Ideology level**: The baseline ideological orientation (Very Liberal to Very Conservative) which dictates how the agent processes the *Source-Stance Congruence*.

## 5. Simulation Workflow

The simulation follows a structured pipeline consisting of six primary phases: Population initialization, information seeding, stochastic social interaction, intervention injection, cognitive reflection & updating, and multi-level metric evaluation.




The 30 agents would be in all experimental conditions together rather than being divided across them. Each condition —- Control, Partisan Refutation, Neutral Institutional, and AI-Generated —- is run as a separate, fully isolated simulation instance using the same agent pool. This winin-agent design ensures that observed differences across conditions reflect the effect of different intervention rather than population sampling variability. Due to the stochastic nature of LLM outputs, each condition is repeated across multiple simulation runs, and means would be reported as the results.

## 6. System Design

The experimental framework is operated by leveraging the **Mesa** library in the Python environment. It adopts a modular architecture consisting of two primary components: the **Dynamic Opinion Agent (DOA)** and the **Agent Interaction Simulator (AIS)**. This structure is inspired by the Fake news Propagation Simulation (FPS) framework proposed by Liu et al. (2024).

The **DOA module** handles the micro-level cognitive processes of individual participants.

The **AIS module** manages the macro-level environmental dynamics. It dictates the interaction protocols, determining the frequency and pairing of agents during daily random exchanges. Furthermore, the AIS coordinates system-wide interventions, such as the distribution of official refutations to the agent population. By monitoring the shifts between susceptible, infected, and recovered states, the AIS facilitates the observation of broad propagation trends over the course of the simulation.

The strengths of this modular framework design are as below:

1. **Logical Decoupling and Experimental Flexibility:**

The modular architecture facilitates a clean separation between individual cognitive processes (DOA) and macro-level social dynamics (AIS). This design allows independent reconfiguration of environmental parameters—such as alter intervention frequencies without necessitating modifications to the underlying agent reasoning logic. Therefore, this approach ensures more granular and precise control over experimental variables.

2. **Enhanced Micro-level Explainability:**

Unlike conventional numerical models that are limited to observing quantitative fluctuations, this framework captures the latent "reasoning trajectories" within the DOA module. By scrutinizing the qualitative logs of agents, the researcher can differentiate whether a shift in an agent's belief stems from social conformity (e.g., peer pressure) or the internal logical validity of the provided interventions. This provides a level of semantic richness and interpretability that is absent in traditional agent-based models (ABMs).

*As President Trump warned in his speech at Eagle Pass, Texas in 2024: "This is "Biden migrant crime", because President Joe Biden allows millions of immigrants to come into the U.S. from other countries. These people mostly come from jails, prisons, mental institutions and asylums, they're terrorists" Trump said, adding that jails from all over the world are "emptying out" into the U.S. The safety of American families demands immediate action.*

This misinformation is politically leaning to conservatives.

7. Misinformation Text and Refutation Text Design

7.1 Misinformation Text:

*Public Safety Report: Federal crime statistics have confirmed what many Americans have long suspected — undocumented immigrants commit violent crimes at nearly twice the rate of native-born citizens, representing an escalating threat to national security. The safety of American families is at serious and growing risk.*

**7.2 Intervention Refutation Text:**

The AIS introduces a new agent designed to represent an official spokesperson. The official agent will issue official refutation with different source to all other agents on designated days to combat the propagation of fake news:

- Authoritative media's rumour refutation notice and explanation with different political stance
    - Liberal source:

"The New York Times, a liberal-leaning major news outlet, has published a fact-check report concluding that the claim regarding undocumented immigrant crime rates being twice those of native-born citizens is misleading and not supported by available evidence. According to peer-reviewed criminological research and official federal crime data, undocumented immigrants have substantially lower crime rates than native-born citizens across a range of felony offenses between 2012 and 2018: Relative to undocumented immigrants, U.S.-born citizens are over 2 times more likely to be arrested for violent crimes, 2.5 times more likely to be arrested for drug crimes, and over 4 times more likely to be arrested for property crimes. Recent federal data also reveals that the incarceration rate for native-born citizens (1,195 per 100,000) is significantly higher than that for undocumented immigrants (674 per 100,000). The New York Times concludes that the claim framing undocumented immigrants as a disproportionate public safety threat is not supported by available evidence. "

    - Conservative source:

"The Wall Street Journal, a conservative-leaning major news outlet, has published a fact-check report concluding that the claim regarding undocumented immigrant crime rates being twice those of native-born citizens is misleading and not supported by available evidence. According to peer-reviewed criminological research and official federal crime data, undocumented immigrants have substantially lower crime rates

than native-born citizens across a range of felony offenses between 2012 and 2018: Relative to undocumented immigrants, U.S.-born citizens are over 2 times more likely to be arrested for violent crimes, 2.5 times more likely to be arrested for drug crimes, and over 4 times more likely to be arrested for property crimes. Recent federal data also reveals that the incarceration rate for native-born citizens (1,195 per 100,000) is significantly higher than that for undocumented immigrants (674 per 100,000). The Wall Street Journal concludes that the claim framing undocumented immigrants as a disproportionate public safety threat is not supported by available evidence. ”

- Authoritative third-party fack checker’s rumour refutation notice and explanation with neutral political stance:

  “PolitiFact, an independent non-partisan fact-checking organization, has published a fact-check report concluding that the claim regarding undocumented immigrant crime rates being twice those of native-born citizens is misleading and not supported by available evidence. According to peer-reviewed criminological research and official federal crime data, undocumented immigrants have substantially lower crime rates than native-born citizens across a range of felony offenses between 2012 and 2018: Relative to undocumented immigrants, U.S.-born citizens are over 2 times more likely to be arrested for violent crimes, 2.5 times more likely to be arrested for drug crimes, and over 4 times more likely to be arrested for property crimes. Recent federal data also reveals that the incarceration rate for native-born citizens (1,195 per 100,000) is significantly higher than that for undocumented immigrants (674 per 100,000). PolitiFact concludes that the claim framing undocumented immigrants as a disproportionate public safety threat is not supported by available evidence. ”

- AI generated rumour refutation notice and explanation:

  “An AI-generated analysis, produced by systematically processing multiple peer-reviewed studies and government datasets without human editorial input or institutional affiliation, concludes that the claim regarding undocumented immigrant crime rates being twice those of native-born citizens is misleading and not supported by available evidence. According to peer-reviewed criminological research and official federal crime data, undocumented immigrants have substantially lower crime rates than native-born citizens across a range of felony offenses between 2012 and 2018: Relative to undocumented immigrants, U.S.-born citizens are over 2 times more likely to be arrested for violent crimes, 2.5 times more likely to be arrested for

drug crimes, and over 4 times more likely to be arrested for property crimes. Recent federal data also reveals that the incarceration rate for native-born citizens (1,195 per 100,000) is significantly higher than that for undocumented immigrants (674 per 100,000). This AI-generated analysis concludes that the claim framing undocumented immigrants as a disproportionate public safety threat is not supported by available evidence. ”